

Lesson Title: ARP, DNS, DHCP, and the Transport Layer

Lesson Objectives / Outcomes

By the end of this lesson, students should be able to:

1. Identify the roles of **ARP**, **DNS**, **DHCP**, and the **Transport Layer** in networking.
2. Explain how IP addresses are assigned **manually** and **automatically**.
3. Understand how these protocols work together to enable device communication.

I. Introduction

Every device in a network needs a **unique IP address** to communicate.

Protocols like **ARP**, **DNS**, and **DHCP** help manage addressing and name resolution, while the **Transport Layer** ensures data is delivered correctly.

II. ARP (Address Resolution Protocol)

Purpose: To map an IP address to a physical (MAC) address.

- When a device wants to communicate with another device in the same network, it uses ARP to find the destination's MAC address.
- Think of ARP as a translator between **IP addresses** (logical) and **MAC addresses** (physical).

Example:

1. Your PC knows the IP of another device (e.g., 192.168.1.10).
2. It sends an ARP request asking, "Who has 192.168.1.10?"
3. The device with that IP replies with its MAC address.

III. DNS (Domain Name System)

Purpose: Converts domain names (like www.google.com) into IP addresses (like 142.250.190.78).

- Without DNS, users would have to remember long IP addresses.

- DNS works like a **phonebook** for the internet.

Process:

1. You type a website name.
2. The DNS server finds and returns the corresponding IP address.
3. Your device connects using that IP.

Example:

www.emna.com → 192.168.10.5

IV. DHCP (Dynamic Host Configuration Protocol)

Purpose: Automatically assigns IP addresses to devices in a network.

How DHCP Works:

1. **Discovery:** The device sends a request asking for an IP.
2. **Offer:** The DHCP server offers an available IP.
3. **Request:** The device requests to use that IP.
4. **Acknowledge:** The server confirms the lease.

Advantages:

- Saves time (no need for manual setup).
- Avoids IP conflicts.
- Automatically configures gateway and DNS settings.

V. Manual vs. Automatic IP Configuration

Method	Description	Example Use
Manual (Static)	The user sets IP address, subnet mask, gateway, and DNS manually.	Servers, printers, routers
Automatic (Dynamic)	IP is assigned automatically by DHCP.	Laptops, phones, tablets

Static IP Setup (Manual):

Example settings on a PC:

- IP: 192.168.1.5
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.1.1
- DNS: 8.8.8.8

Dynamic IP Setup (Automatic):

- Enable “Obtain an IP address automatically” in the network settings.
- The DHCP server provides all needed details.

VI. Transport Layer Overview

Purpose: Ensures data is delivered reliably between devices.

- It works between the **Application Layer** and **Network Layer**.
- Responsible for **error checking, data sequencing, and retransmission**.

Main Protocols:

- **TCP (Transmission Control Protocol):** Reliable, connection-oriented (e.g., web browsing, email).
- **UDP (User Datagram Protocol):** Faster but connectionless (e.g., streaming, gaming).

VII. How They Work Together

1. **DHCP** gives your device an IP address.
2. **DNS** converts website names into IP addresses.
3. **ARP** finds the MAC address to send data locally.
4. **Transport Layer (TCP/UDP)** ensures the data arrives correctly.

Together, they make **network communication smooth and automatic**.

VIII. Summary

- **ARP:** Finds MAC address from IP.
- **DNS:** Translates domain names to IPs.
- **DHCP:** Automatically assigns IP addresses.
- **Transport Layer:** Ensures reliable data delivery.
- IP addresses can be given **manually (static)** or **automatically (DHCP)** depending on the need.

IX. Short Activity (15–20 minutes)

Instruction:

Each student must do one of the following:

1. Write the steps to manually set an IP address on a PC or phone.
2. Observe and note the current IP configuration on their device (using ipconfig or Wi-Fi settings).
3. Identify whether their device uses DHCP or Static IP and explain why.